
CIS437 - Cloud Computing

– From Computing History to Modern Cloud Platforms –

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Last Class

- Google Cloud Console
- Credits
- Projects
- Virtual Machines
- Assignment 1

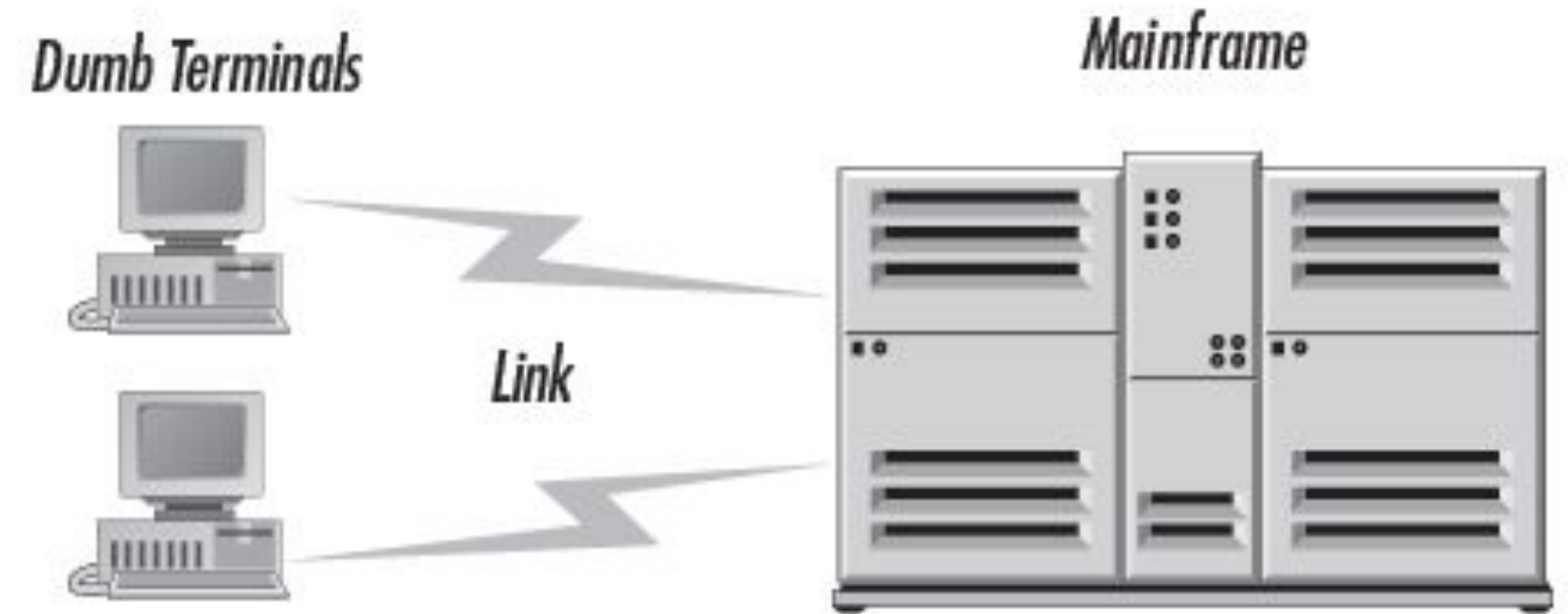
Agenda

- From centralized computing to cloud computing
- Virtualization and containers
- What is a cloud platform?
- Google Cloud services and interfaces
- Cloud platform trade-offs

From mainframes to cloud

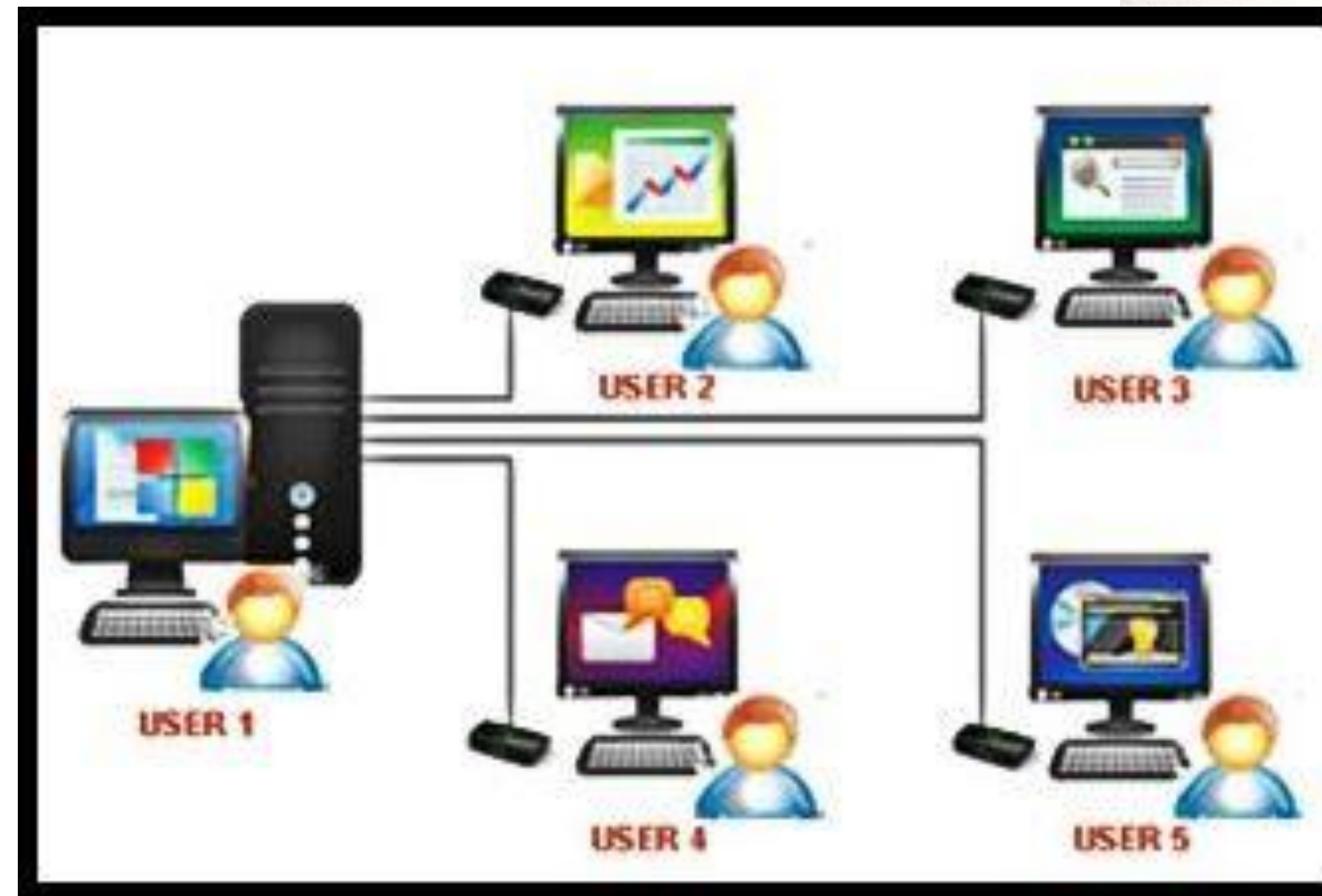
Initially, many users accessed a centralized mainframe through terminals with little or no local processing.

- Most computation occurred on the central system
- Users shared expensive computing resources
- Sound familiar?

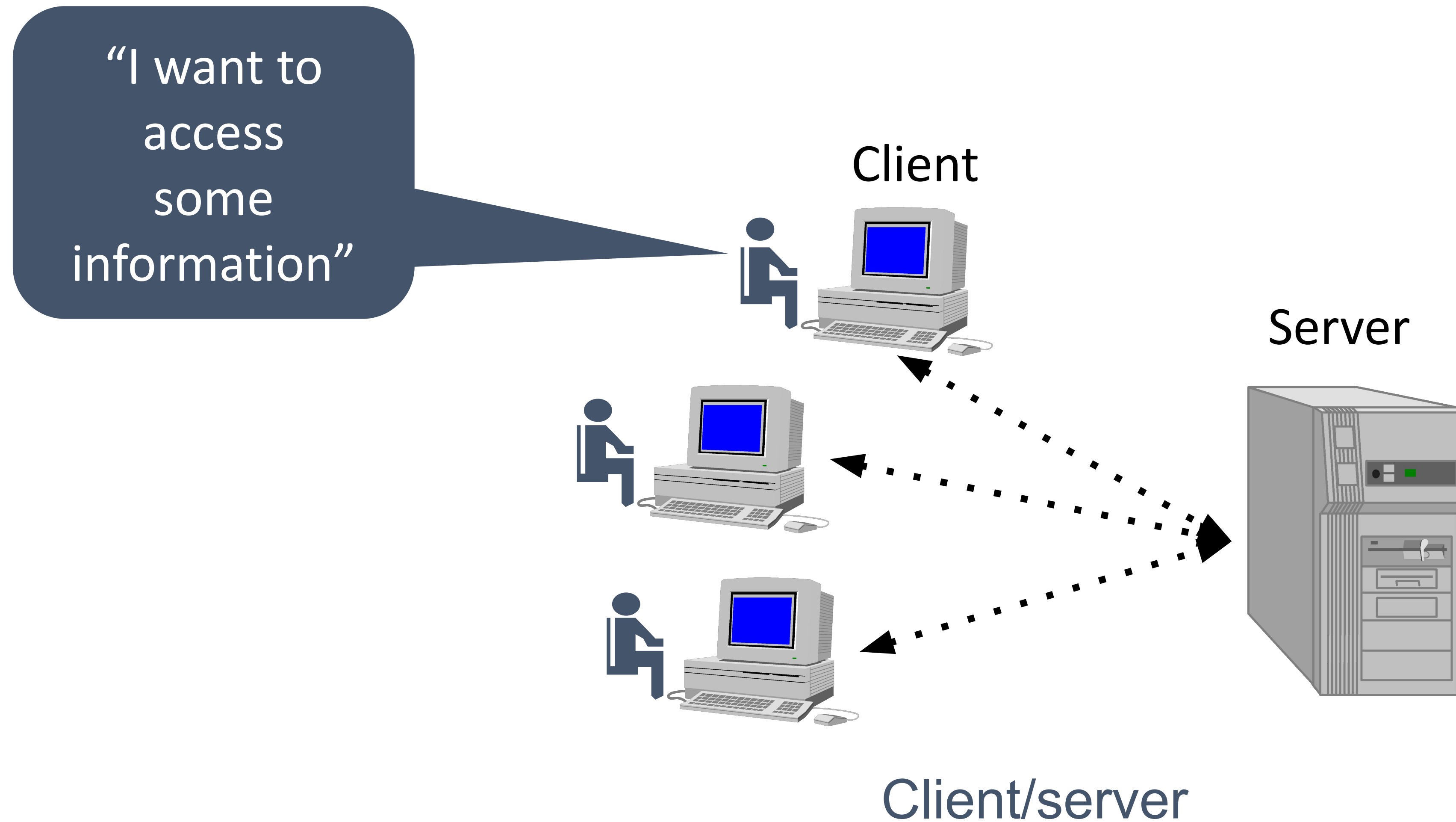


Time Sharing and Personal Computing

- Multiple users could access a system
- **Time sharing:** multiple programs share CPU time
- Operating systems manage resource allocation
- Computing later shifted toward PCs and client/server systems



Client/server Architecture



How Client/Server Works

Client

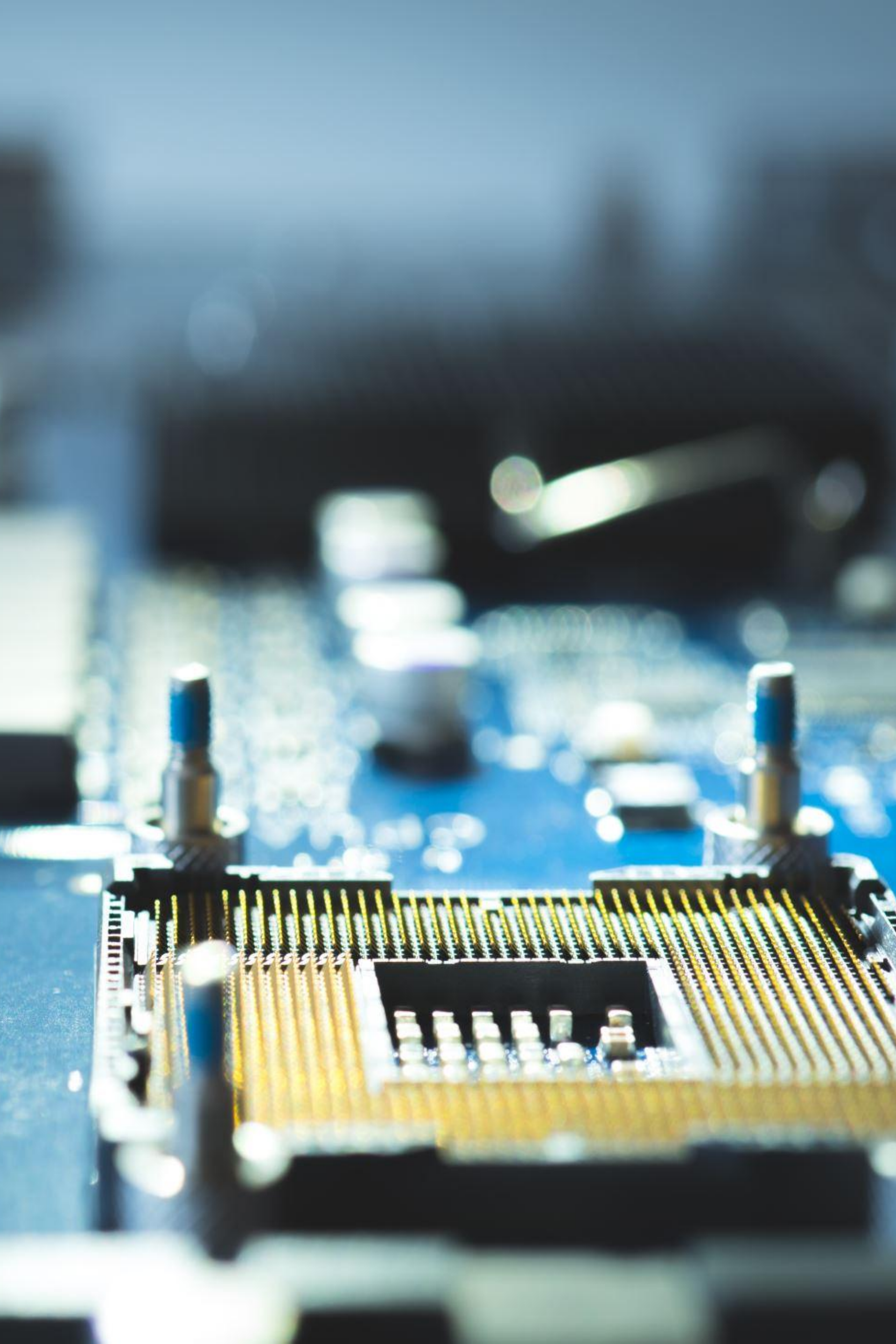
- Initiates a request
- Needs to know how to reach the server

Server

- Waits for incoming requests
- Processes requests and returns responses

Client/Server → Limitations

- Scaling requires additional capacity
- A single server may become a single point of failure
- Servers require administration and maintenance
- Capacity must be planned in advance
- Overprovisioning can leave resources underutilized



Virtualization: Many VMs on Shared Hardware

Run multiple machines / operating systems on a single system

- Independently and concurrently
- Multiple users each log into their own VM

Why virtualization?

Consider an enterprise data center

- Always need more servers!
- Easier to virtualize workstations/servers
- Cost-effective
- Less hardware to deal with
- Easy to configure and deploy

How Virtualization Works: The Hypervisor

- Each VM behaves as if it controls its own hardware
- A **hypervisor** manages access to physical CPU, memory, storage, and devices
- The hypervisor isolates VMs from one another
- This creates an abstraction between physical hardware and virtual machines

Virtual Machines vs. Containers

VM	Container
Includes guest OS	Shares host kernel
Heavier	Lighter
Typically Slower startup	Typically faster startup
Strong machine abstraction	Application/process-level isolation



From Virtualization to Cloud Platforms

Physical hardware



Virtualization



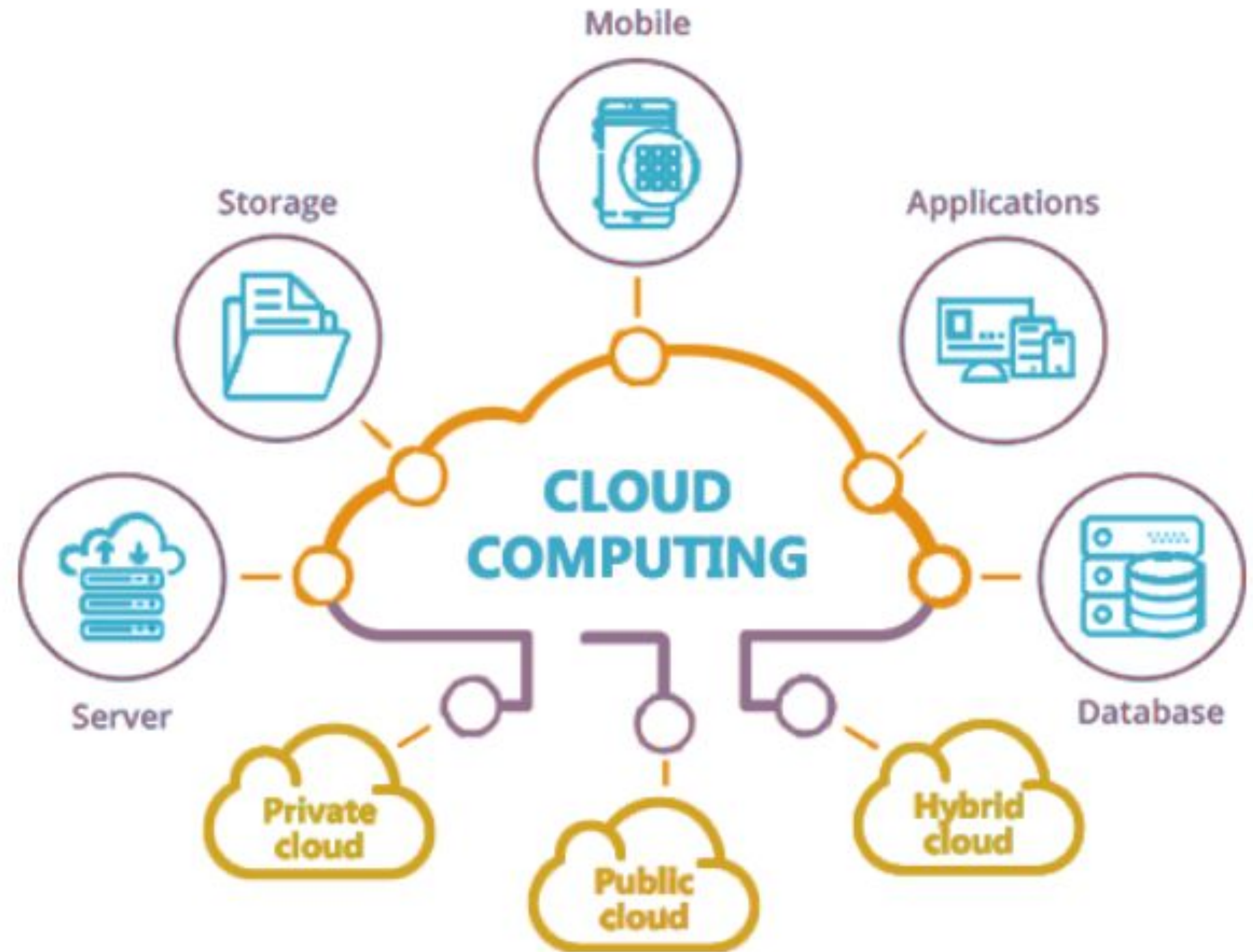
Virtual machines / containers



Automation + APIs + networking + storage



Cloud platform



What Is a Cloud Platform?

A cloud platform provides computing resources and managed services on demand through software interfaces and APIs.

Compute | Storage | Networking | Databases | Security | Managed Services

Examples of Cloud Platforms

Major public cloud platforms

- Google Cloud
- Amazon Web Services (AWS)
- Microsoft Azure
- Oracle Cloud
- IBM Cloud

Open-source/private-cloud platforms

- OpenStack
- OpenNebula

Source: <https://www.softwaretestinghelp.com/cloud-computing-service-providers/>

One Task, Multiple Interfaces

- Create a VM
 - Google Cloud Console
 - gcloud CLI
 - REST API / SDK
 - Infrastructure as Code

`gcloud compute instances create my-vm`

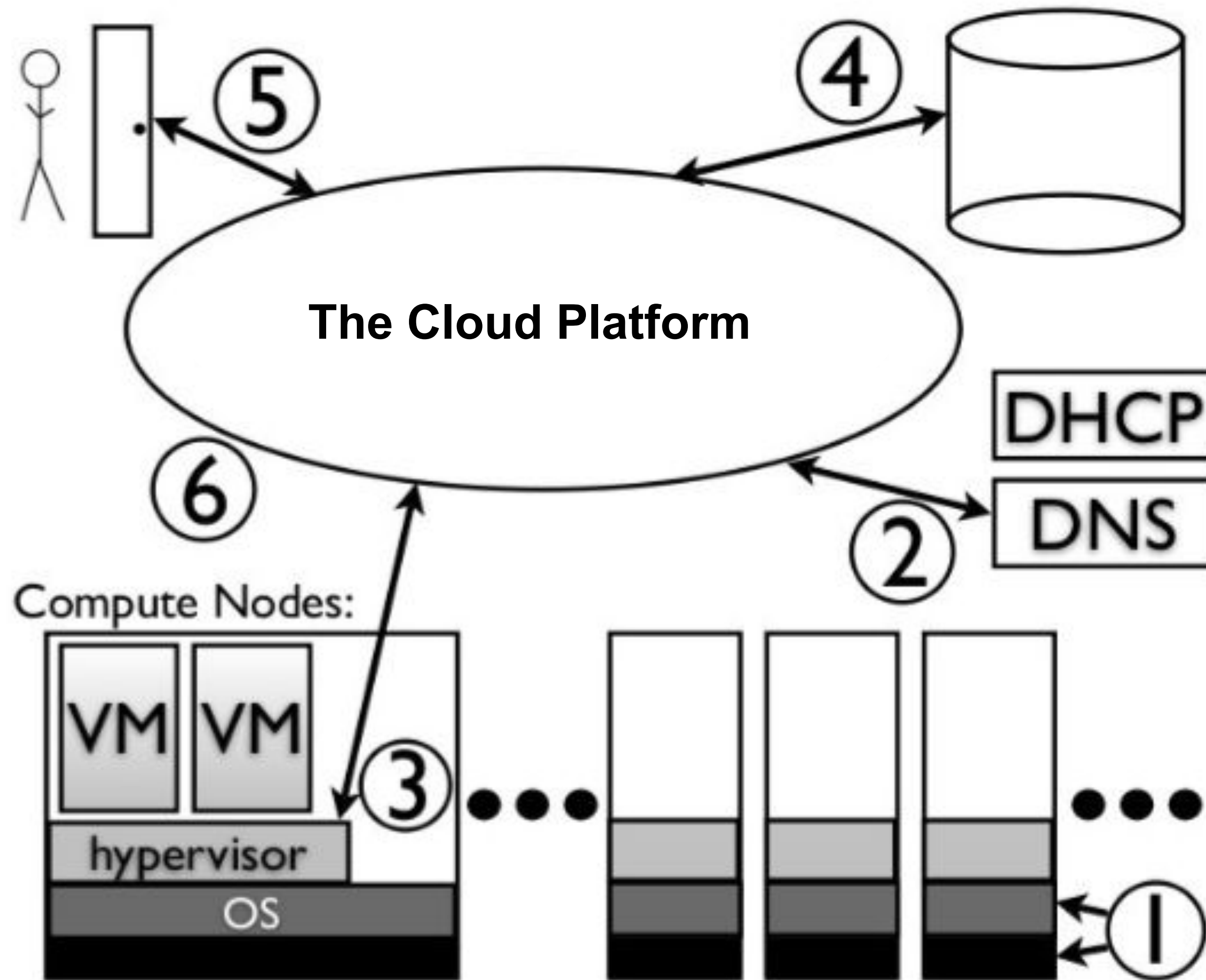
Major Google Cloud Service Families

Need	Google Cloud examples
Compute	Compute Engine
Storage	Cloud Storage
Data & Databases	Cloud SQL, Firestore, BigQuery
Networking	VPC, Load Balancing
Containers	GKE, Cloud Run
Serverless	Cloud Run functions
Security	IAM, Secret Manager
Operations	Cloud Monitoring, Logging
AI/ML	Vertex AI

What Should a Cloud Platform Provide?

- Availability and geographic choice
- Scalability and elasticity
- Reliability and fault tolerance
- Networking
- Identity and access management
- Monitoring and logging
- APIs and automation
- Security
- Pricing and cost management

Behind the Cloud Platform



1. Select an OS image
2. Allocate compute resources
3. Place the VM on physical infrastructure
4. Attach virtual storage
5. Configure virtual networking and IP addresses
6. Expose management through the cloud platform

Ways to Use Google Cloud

Console

→ visual/manual

Cloud Shell + **gcloud**

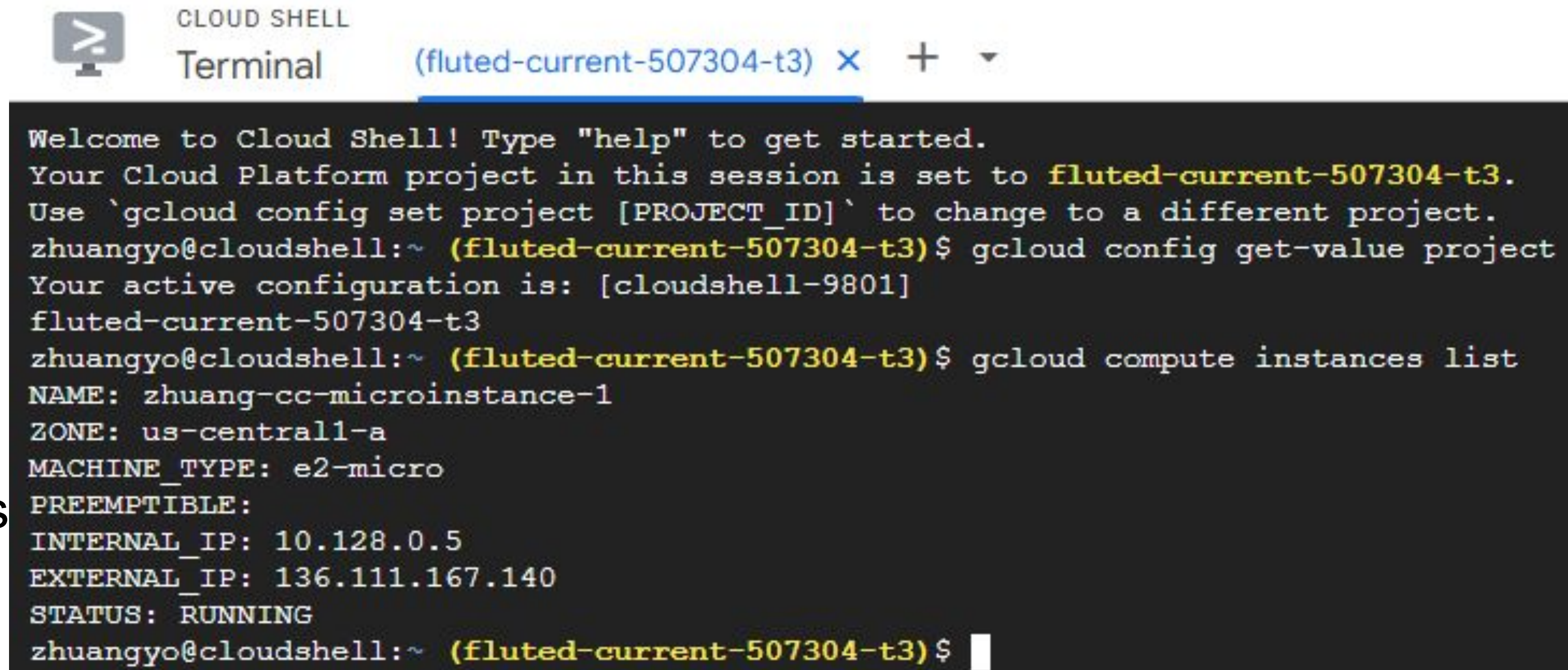
→ command line

APIs / SDKs

→ application/programmatic access

Infrastructure as Code

→ repeatable infrastructure



```
CLOUD SHELL
Terminal (fluted-current-507304-t3) x + ▾

Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to fluted-current-507304-t3.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud config get-value project
Your active configuration is: [cloudshell-9801]
fluted-current-507304-t3
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud compute instances list
NAME: zhuang-cc-microinstance-1
ZONE: us-central1-a
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.5
EXTERNAL_IP: 136.111.167.140
STATUS: RUNNING
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$
```

```
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud projects describe fluted-current-507304-t3
createTime: '2026-09-01T04:09:41.837Z'
lifecycleState: ACTIVE
name: zhuang-CloudAppDev
parent:
  id: '738789643771'
  type: organization
projectId: fluted-current-507304-t3
projectNumber: '719382803826'
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$
```

Cloud console

Cloud Platforms Involve Trade-offs / Advantages

Advantages of Cloud Computing

- Elasticity
- Rapid provisioning
- Reduced upfront hardware investment
- Access to managed services
- Global infrastructure
- Potential cost efficiency
- Potential reliability with good design



Cloud Platforms Involve Trade-offs / Disadvantages

Disadvantages of Cloud Computing

- Security, privacy, and compliance
- Cost complexity
- Vendor lock-in
- Required cloud expertise
- Network dependency and latency
- Governance and organizational complexity

